Cyprus University of Technology

CSE 226
Algorithms and Complexity

Lecture 21
NP-Completeness (Part 2)
NP-Πληρότητα (Μέρος 2)

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Review: **P**, **NP**, **NPC**

- **P** = the set of problems that can be *solved* in polynomial time
- **NP** = the set of problems for which a solution can be *verified* in polynomial time
- **NP-Complete** = the "hardest" problems in NP
  - No polynomial-time algorithm has been discovered for an NP-Complete problem
  - No proof that a polynomial-time algorithm cannot exist for any of the NP-Complete problems
- Any problem in **P** is also in **NP**
- Unknown whether **P** = **NP** (most suspect not)

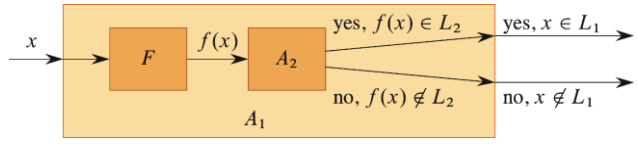
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Polynomial Reductions

- A polynomial **reduction** (**αναγωγή**) from the decision problem Q_1 to the decision problem Q_2 is a function f that satisfies the two conditions:
  - f is computable by a polynomial time algorithm, and
  - for any instance I of Q_1 the answer is "yes" if and only if for the instance $f(I)$ of Q_2 the answer is "yes"
- We write: $Q_1 \leq_P Q_2$ (Q_1 reduces to Q_2)
- Intuitively: If Q_1 reduces in polynomial time to Q_2, Q_1 is "no harder to solve" than Q_2

Polynomial Reductions

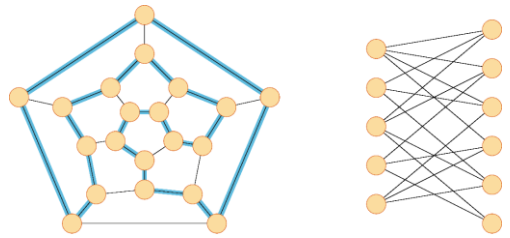
- **Theorem:** If $Q_1 \leq_P Q_2$ and $Q_2 \in \mathbf{P}$, then $Q_1 \in \mathbf{P}$
- Proof:



- **Equivalently:** if $Q_1 \leq_P Q_2$ and $Q_1 \notin \mathbf{P}$, then $Q_2 \notin \mathbf{P}$
- **Theorem**: If $Q_1 \leq_P Q_2$ and $Q_2 \leq_P Q_3$, then $Q_1 \leq_P Q_3$
- Two problems Q_1 and Q_2 are called **polynomially equivalent** (**πολυωνυμικά ισοδύναμα**) whenever both $Q_1 \leq_P Q_2$ and $Q_2 \leq_P Q_1$

Example: Hamiltonian Cycles (HC)

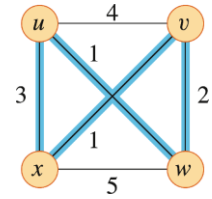
- A **simple cycle** (**απλός κύκλος**) in a graph $G = (V, E)$ is a sequence $\langle v_1, v_2, \ldots, v_k \rangle$ of distinct vertices from V such that $(v_i, v_{i+1}) \in E$ for all $i, 1 \leq i \leq k$ and $(v_k, v_1) \in E$
- A **hamiltonian cycle** of an undirected graph is a simple cycle that contains every vertex
- The **hamiltonian-cycle problem**: Given a graph G, does it have a hamiltonian cycle?



Example: Traveling-Salesman Problem (TSP)

- **Traveling-salesman problem**: Given a set of cities $C = \{c_1, c_2, \ldots, c_m\}$ and the distance $d(c_i, c_j)$ between each pair of cities c_i, c_j, find an ordering of these cities $\langle c_{\pi(1)}, c_{\pi(2)}, \ldots, c_{\pi(m)} \rangle$ that minimizes the total distance

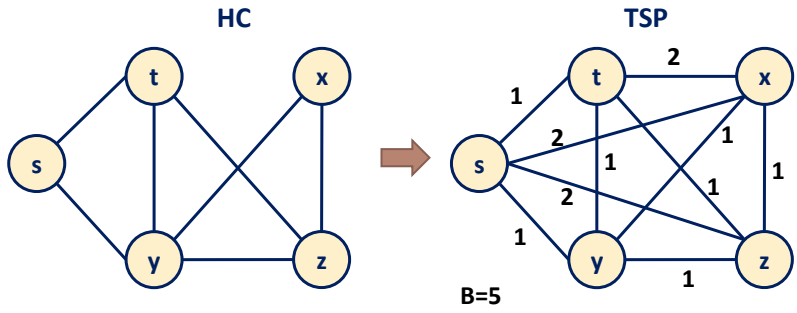
$$\left[\sum_{i=1}^{m-1} d(c_{\pi(i)}, c_{\pi(i+1)})\right] + d(c_{\pi(m)}, c_{\pi(1)})$$



- Equivalent decision problem:
  - …, is there an ordering of these cities (tour) such that the total distance is at most B?

Example: Reduce HC to TSP

1. Find a transformation
   - Let the vertices of G of HC denote the cities of TSP
   - Define the distance between cities as $d(v_i, v_j) = 1$ if $(v_i, v_j) \in E$, and 2 otherwise
   - The bound B on the desired tour length is $|V| = m$



Example: Reduce HC to TSP

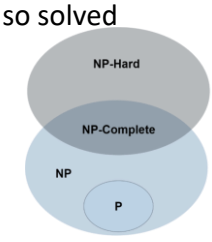
1. Find a transformation
   - Let the vertices of G of HC denote the cities of TSP
   - Define the distance between cities as $d(v_i, v_j) = 1$ if $(v_i, v_j) \in E$, and 2 otherwise
   - The bound B on the desired tour length is $|V| = m$
2. Prove that it can be done in polynomial time
   - There are $m \times (m-1)/2$ distances $d(v_i, v_j)$
3. Prove equivalency
   - ($\Rightarrow$) Suppose that $\langle v_1, v_2, \ldots, v_m \rangle$ is a Hamiltonian Cycle in G
   - Then, $\langle v_1, v_2, \ldots, v_m \rangle$ is also a tour in $f(G)$, and this tour has total length $m = B$ because each intercity distance travelled in the tour corresponds to an edge of G and hence has length 1

Example: Reduce HC to TSP

3. Prove equivalency (cont'd)
   - (⇐) Suppose that $\langle v_1, v_2, \ldots, v_m \rangle$ is a tour in $f(G)$ with total length no more than B
   - The fact that $B = m$ implies that each pair of successively visited cities must be exactly distance 1 apart
     - because any two cities are either distance 1 or distance 2 apart, and
     - exactly m such distances are summed in computing the tour length
   - By the definition of $f(G)$, it follows that (v_i, v_{i+1}) and (v_m, v_1) are all edges of G, and hence $\langle v_1, v_2, \ldots, v_m \rangle$ is a Hamiltonian cycle for G
- **Conclusion:** HC $\leq_P$ TSP
  - If TSP can be solved by a polynomial time algorithm then so can HC
  - Equivalently, if HC is proved intractable, then so is TSP

NP-Completeness

- A problem Q is **NP-Complete** if
  1. $Q \in \mathbf{NP}$
  2. $Q' \leq_P Q$ for all other decision problems $Q' \in \mathbf{NP}$
- Q is **NP-Hard** if the first condition is not necessarily true
- Therefore, the **NP-Complete** problems are "the hardest problems in **NP**"
  - If any single **NP-Complete** problem can be solved in polynomial time, then all problems in **NP** can be so solved
  - If any problem in **NP** is intractable, then so are all **NP-Complete** problems
- Opinion of most theoretical comp. scientists:
  - $\mathbf{P} \subset \mathbf{NP}$, $\mathbf{NPC} \subset \mathbf{NP}$, $\mathbf{P} \cap \mathbf{NPC} = \emptyset$



NP-Completeness

- **Note:** If $Q_1, Q_2 \in \mathbf{NP}$, $Q_1 \in \mathbf{NPC}$, and $Q_1 \leq_P Q_2$, then $Q_2 \in \mathbf{NPC}$
- **Idea:** To prove that a new problem Q is **NP-Complete**, show that:
  - $Q \in \mathbf{NP}$, and
  - some known **NP-Complete** problem Q' reduces to Q
    - Select problem Q'
    - Find a transformation f from Q' to Q
    - Prove that f is a polynomial reduction
- Problem: we need a *first* **NP-Complete** problem
  - Cook comes to a rescue

The Satisfiability Problem (SAT)

- Consider a set of boolean variables: $U = \{u_1, u_2, \ldots, u_m\}$
- A **truth assignment** for U is a function $t : U \to \{\text{True}, \text{False}\}$
- For a variable $u \in U$, u and $\neg u$ are **literals** (**λέκτημα**) over U
- A **clause** (**φράση**) C over U is a **disjunction** (**διάζευξη**) of literals over U, e.g., $u_1 \vee \neg u_3 \vee u_8$
- A **satisfying assignment** is a truth assignment that causes the clause to evaluate to True
- A **formula** Φ is a **conjunction** (**σύζευξη**) of clauses $(u_1 \vee \neg u_3 \vee u_8) \wedge (u_7) \wedge (\neg u_4 \vee \neg u_1)$

The Satisfiability Problem (SAT)

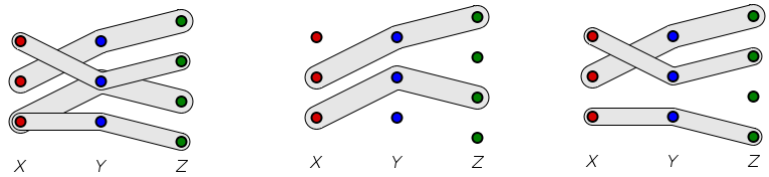
- A formula Φ is **satisfiable** if and only if there is a truth assignment that simultaneously satisfies all the clauses $(u_1 \vee \neg u_3 \vee u_8) \wedge (u_7) \wedge (\neg u_4 \vee \neg u_1)$
- The **Satisfiability problem** (**SAT**): is a given formula satisfiable?
- **Cook's Theorem**: SAT is **NP-Complete**
- We need to be aware of known **NP-Complete** problems
  - We'll cover now six more (Karp, 1972)

Six Basic NP-Complete Problems (1)

- **3-Satisfiability** (**3SAT**)
  - INSTANCE: Collection $C = \{c_1, c_2, \ldots, c_m\}$ of clauses on a finite set U of variables such that $|c_i| = 3$ for $1 \leq i \leq m$
  - QUESTION: Is there a truth assignment for U that satisfies all the clauses in C?
- Example: $\Phi = (u_1 \vee u_2 \vee \neg u_4) \wedge (\neg u_2 \vee \neg u_3 \vee u_5)$
  - Φ is satisfied if u_1= true and u_2= true and the rest are false
  - Other solutions exist

Six Basic NP-Complete Problems (2)

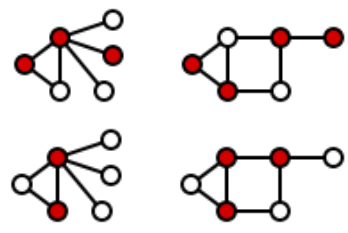
- **3-Dimensional Matching** (**3DM**)
  - INSTANCE: A set $M \subseteq X \times Y \times Z$, where X, Y, Z are disjoint sets (i.e., M is a set of triplets $\langle x \in X, y \in Y, z \in Z \rangle$)
  - QUESTION: Does M contain a *matching*, that is, a subset $M' \subseteq M$ such that $|M'| \geq q$ and no two elements of M' agree in any coordinate?
- Example:



Input A solution Another solution

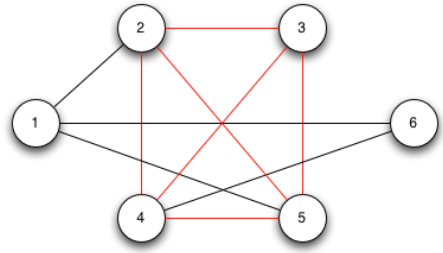
Six Basic NP-Complete Problems (3)

- **Vertex Cover** (**VC**)
  - INSTANCE: A graph $G = (V, E)$ and a positive integer $K \leq |V|$
  - QUESTION: Is there a vertex cover of size at most K for G, that is, a subset $V' \subseteq V$ such that $|V'| \leq K$ and, for each edge $(u, v) \in E$, at least one of u and v belongs to V'
- Example
  - A vertex cover
  - A minimum vertex cover



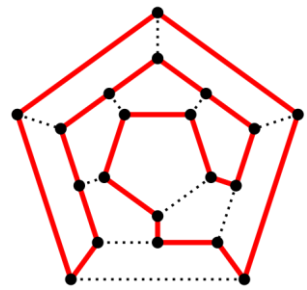
Six Basic NP-Complete Problems (4)

- **Clique**
  - INSTANCE: A graph $G = (V, E)$ and a positive integer $J \leq |V|$
  - QUESTION: Does G contain a clique of size J or more, that is, a subset $V' \subseteq V$ such that $|V'| \geq J$ and every two vertices in V' are joined by an edge in E?
- Example



Six Basic NP-Complete Problems (5)

- **Hamiltonian Cycle** (**HC**)
  - INSTANCE: A graph $G = (V, E)$
  - QUESTION: Does G contain a Hamiltonian cycle, that is, an ordering $\langle v_1, v_2, \ldots, v_k \rangle$ of the vertices of G, where $k = |V|$, such that $(v_k, v_1) \in E$ and $(v_i, v_{i+1}) \in E$ for all $i, 1 \leq i \leq k$?
- Example

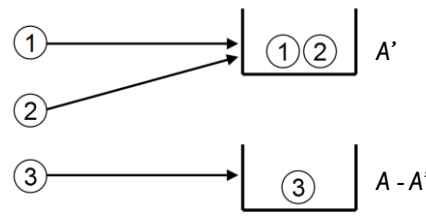


Six Basic NP-Complete Problems (6)

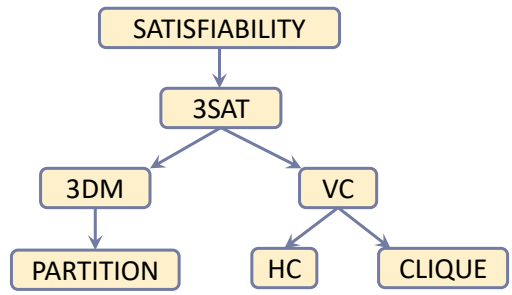
- **Partition**
  - INSTANCE: A finite set A and a "size" $s(a) \in Z^+$ for each $a \in A$
  - QUESTION: Is there a subset $A' \subseteq A$ such that

$$\sum_{a \in A'} s(a) = \sum_{a \in A - A'} s(a)$$

- Example: (here $s(a) = a$)



Sequence of Transformations



- The sequence of transformations is not unique. One may prove **NP**-Completeness by transforming other **NP-Complete** problems
  - Example: Clique → VC, or 3-SAT → Clique